The Pythagorean Theorem

Concept

The Pythagorean theorem connects the three sides of a right triangle.

Hypotenuse

c is opposite the 90° angle and is the longest side.

Formula

$a^2 + b^2 = c^2$, for a right triangle only.

Missing leg

$a = \sqrt{(c^2 - b^2)}$; subtract, then take the positive root.

Check

Use the same units; the hypotenuse must be longest.

Example 1

Find the hypotenuse when the legs are 9 cm and 12 cm.

Solution

1. $c^2 = 9^2 + 12^2 = 81 + 144 = 225$.

2. $c = \sqrt{225} = 15$ cm.

The Pythagorean Theorem

Example 2

A 13 m ladder reaches 12 m up a wall. Find the distance from the wall.

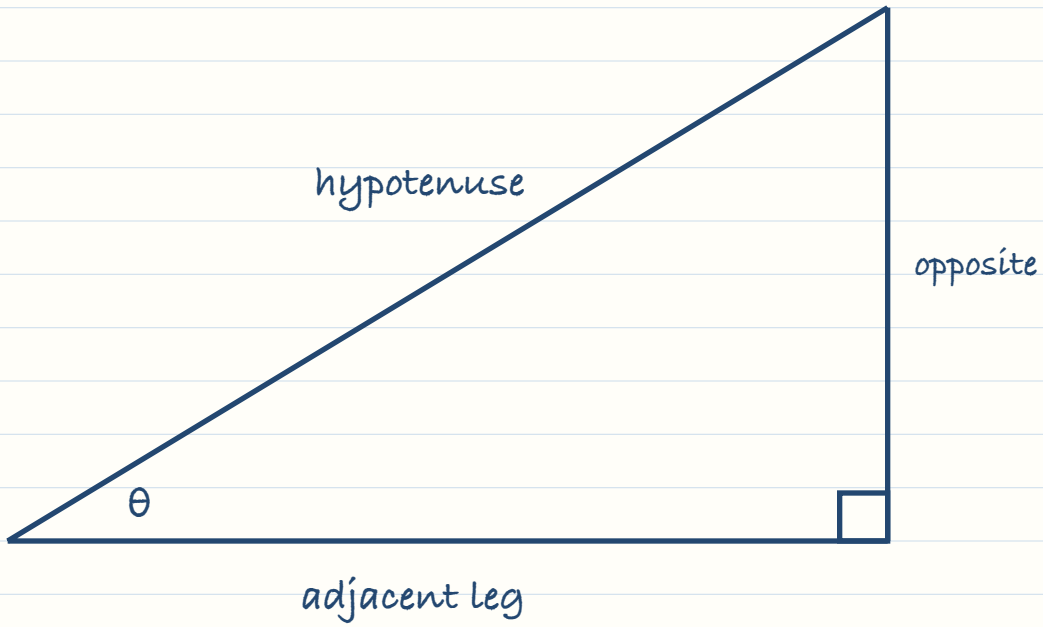
Solution

1. The ladder is the hypotenuse: $x^2 + 12^2 = 13^2$.

2. $x^2 = 169 - 144 = 25 \rightarrow x = 5$ m.

The Pythagorean Theorem

Visual notes | label before calculating



The hypotenuse is opposite the right angle. Opposite and adjacent refer to the selected acute angle θ. A different selected angle changes those two labels.

Here a and b label the two legs and c labels the hypotenuse. Squares built on the legs have total area $a^2 + b^2$, equal to c^2, the area of the square built on the hypotenuse.

The Pythagorean Theorem

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the hypotenuse with legs 5 and 12.

2. Find a leg if the hypotenuse is 10 and the other leg is 6.

3. Is a triangle with sides 8, 15 and 17 right?

4. Find the diagonal of a square with side 7 cm, exactly and to two decimals.

The Pythagorean Theorem

Practice answers | check after trying

1. $\sqrt{(25 + 144)} = 13$.

2. $\sqrt{(100 - 36)} = 8$.

3. Yes: $64 + 225 = 289 = 17^2$.

4. $\sqrt{98} = 7\sqrt{2}$ cm ≈ 9.90 cm. The exact radical avoids rounding error.